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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 29: Introduction to Sustainable Transport Planning

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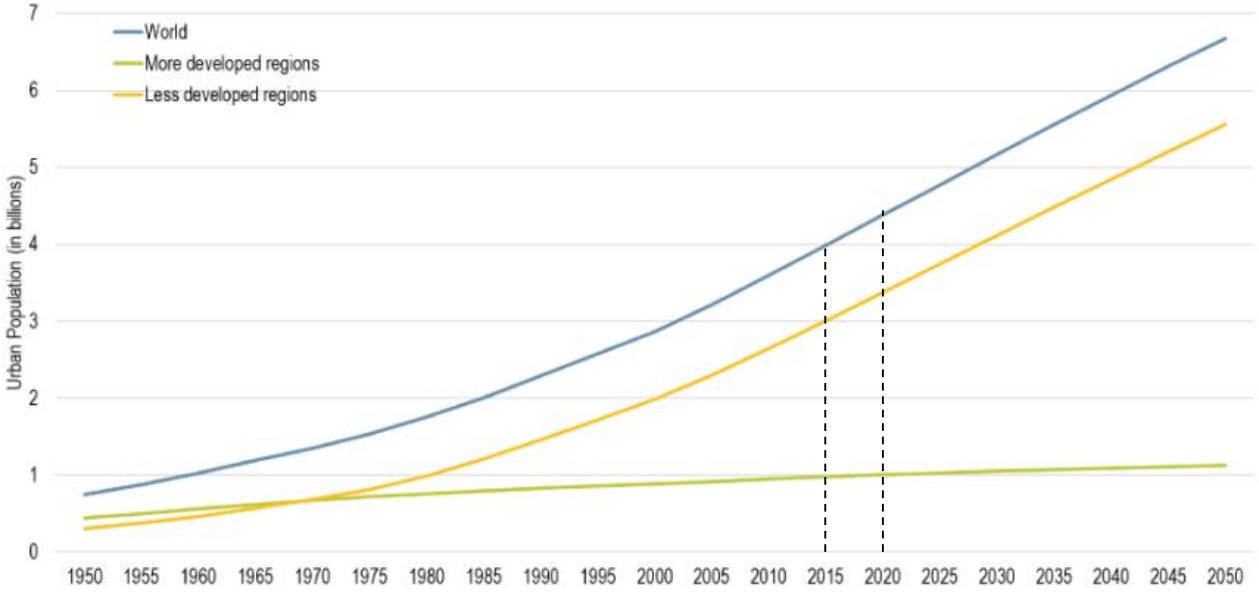


Contents

- Urbanization and the need for Transport Planning
- Problems to be addressed in Planning process
- What are the solutions?
- Conclusion



Urbanization: The main Concern



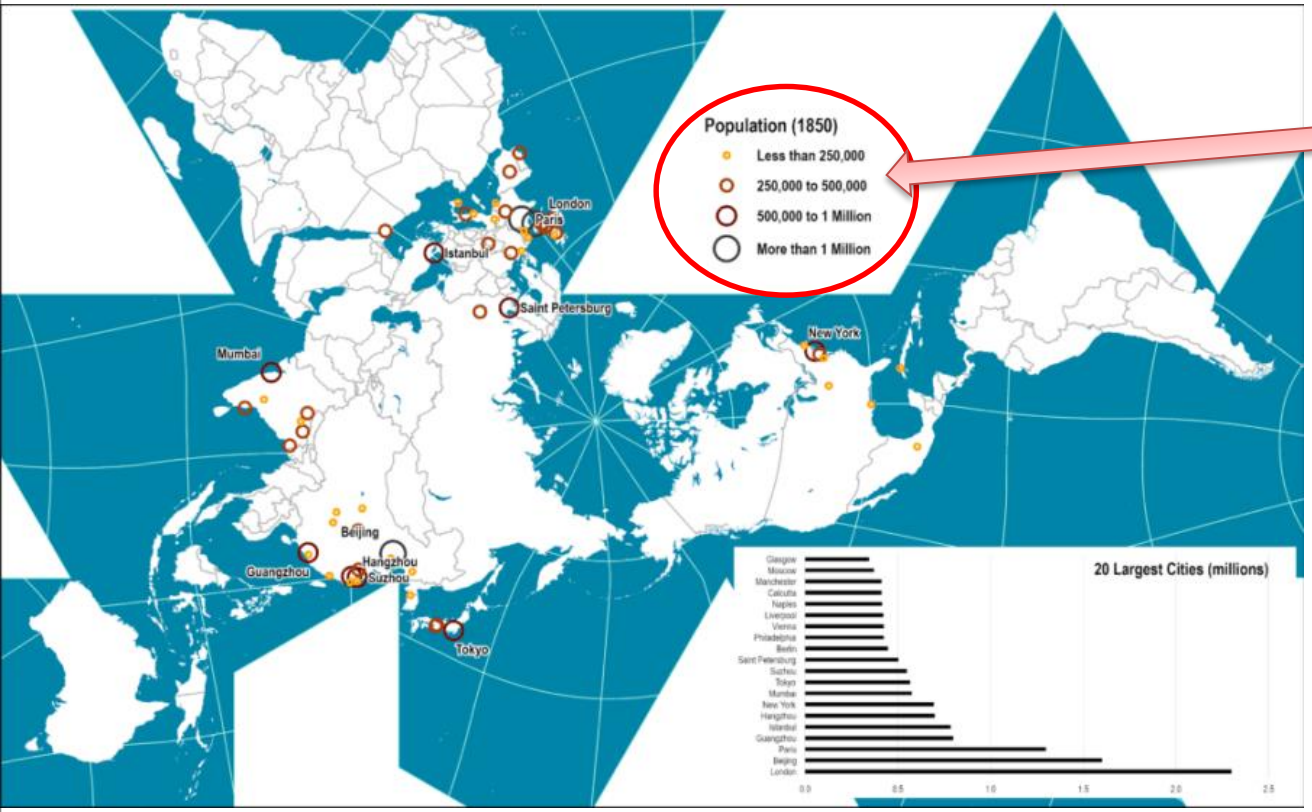
World Urban population 1950-2015 with Projections to 2050

Source: (Jean-Paul Rodrigue 2020)

Global urban population tripled between 1950 and 2015.

By 2050, 70% o the Global population could be urbanized (6.4 billion urban residents

World's largest cities and Population in 1850



- Only 3 cities had population > 1 million.
  - (London, Beijing and Paris)

Source: (Jean-Paul Rodrigue 2020)

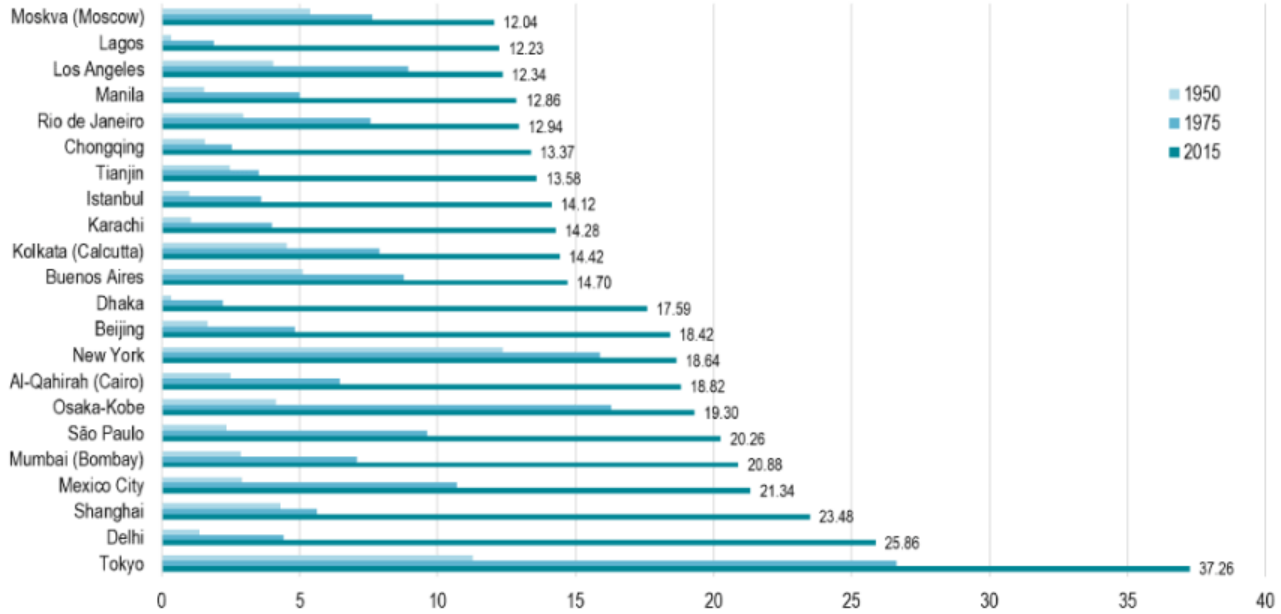
World's largest cities and Population in 2015



Source: (Jean-Paul Rodrigue 2020)

- Emergence of several megacities with population > 8-10 million

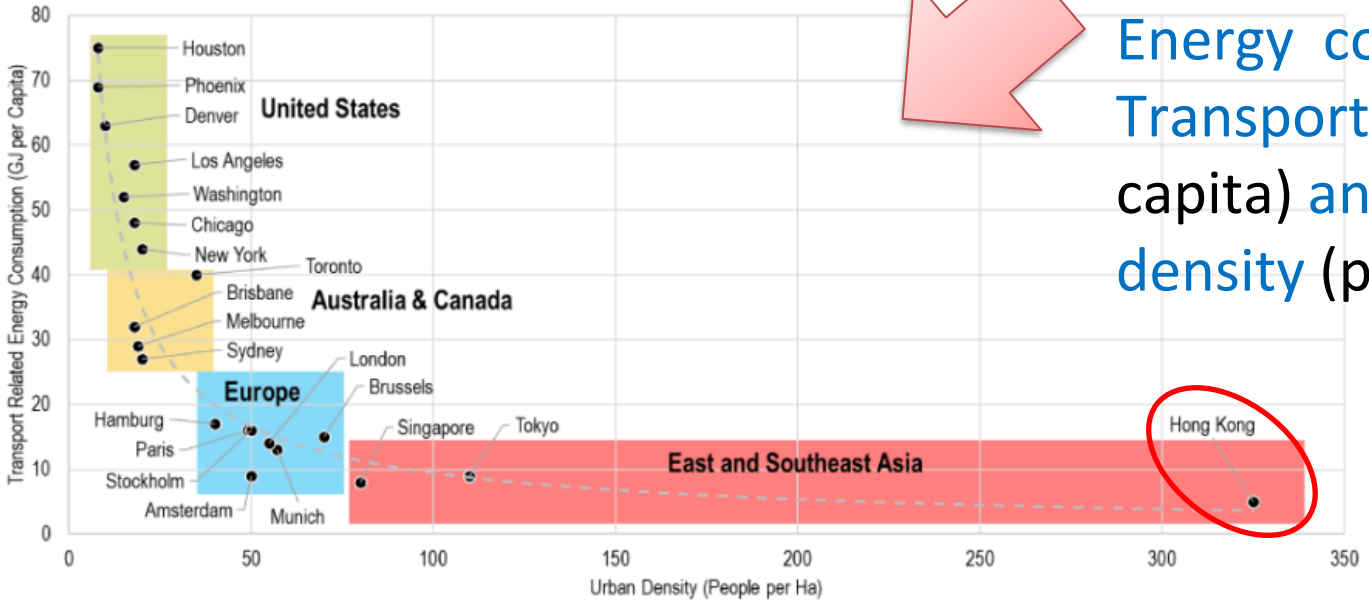
Cities with Population > 12 million in 2015



Year	No. of Metropolitan cities
1950	83
1975	160
2015	514

Source: (Jean-Paul Rodrigue 2020)

Urban Density and Energy Consumption in Transport sector

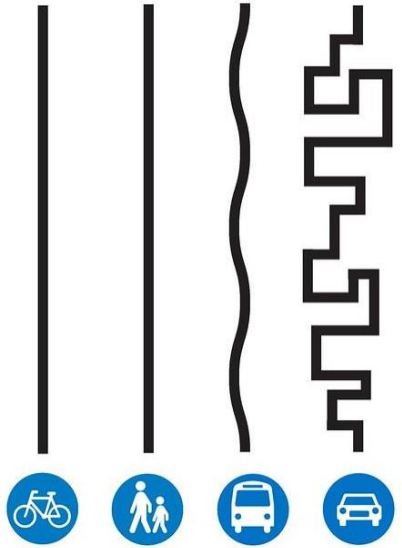


Relationship between the Energy consumption in Transport sector (GJ per capita) and the Urban density (people per Ha)

Source: (Jean-Paul Rodrigue 2020)

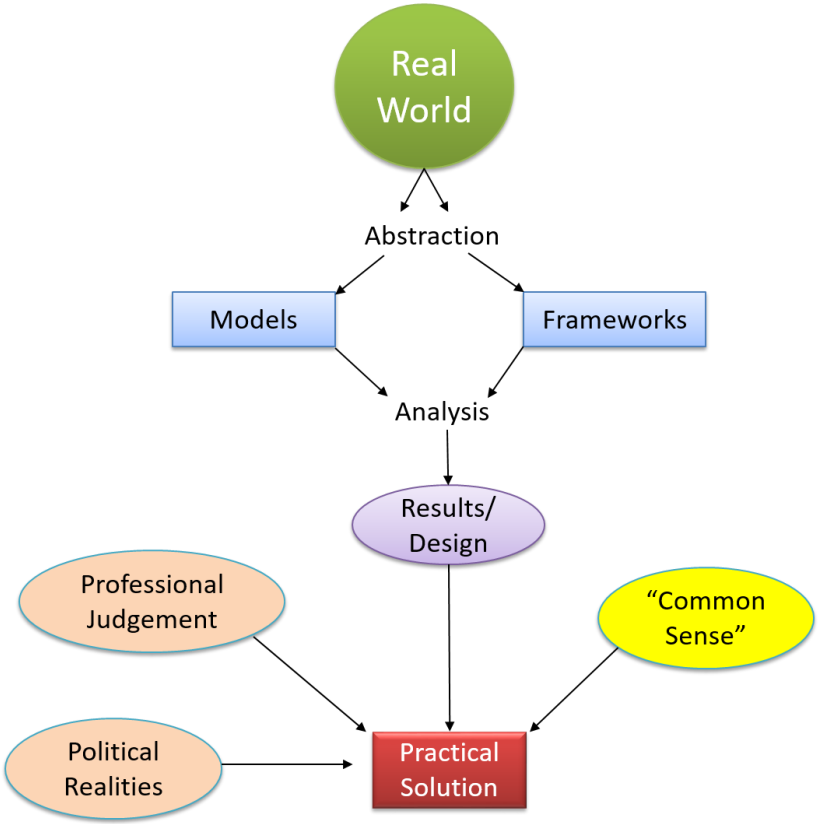
The Need for Transport Planning

- Though Transportation provides significant benefits to humans, there exists many negative externalities.
- Careful and appropriate Transport planning is a necessity in the allocation, design and construction phases of every transport projects along with efficient policy implementations to achieve optimal benefits.
- Transport safety and security along with the Natural and anthropogenic disasters are some major challenges in transport planning.



Source: (Jean-Paul Rodrigue 2020)

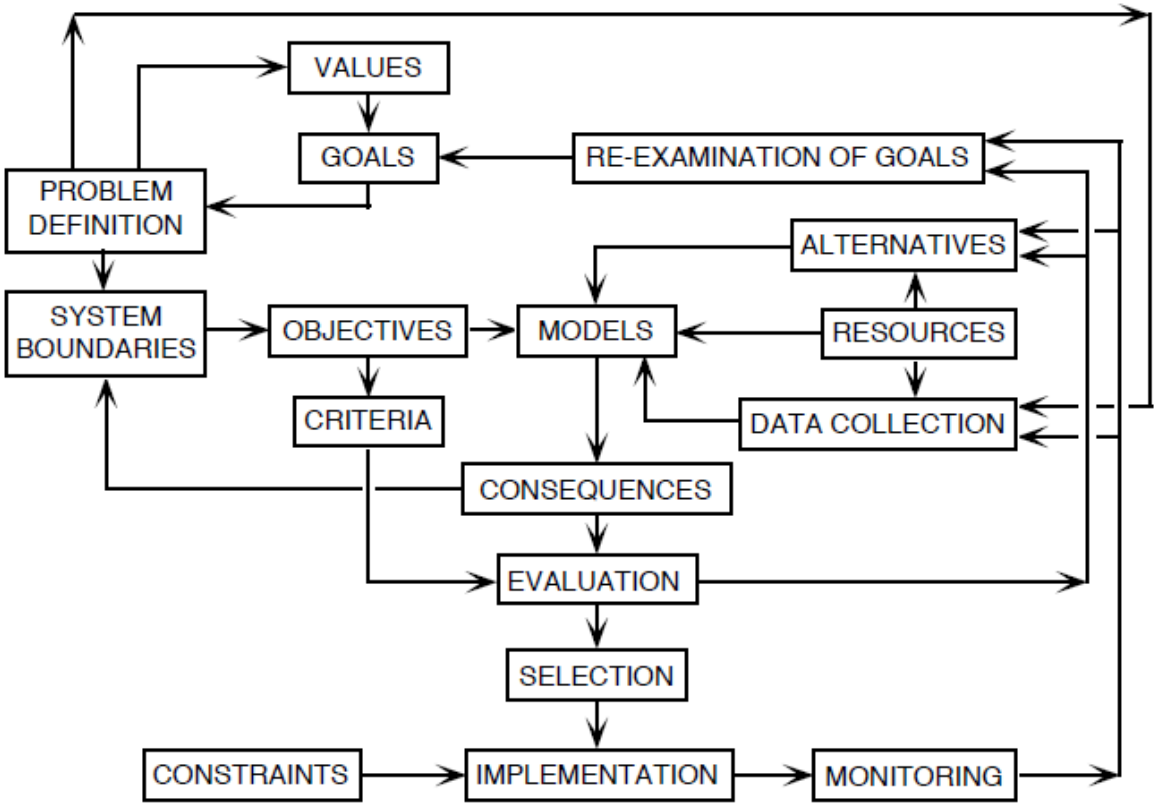
What exactly is Planning?



Source: (MIT lecture Notes, 2002)

Solving the Real World problems with Practical Solutions

The Transportation Planning Process

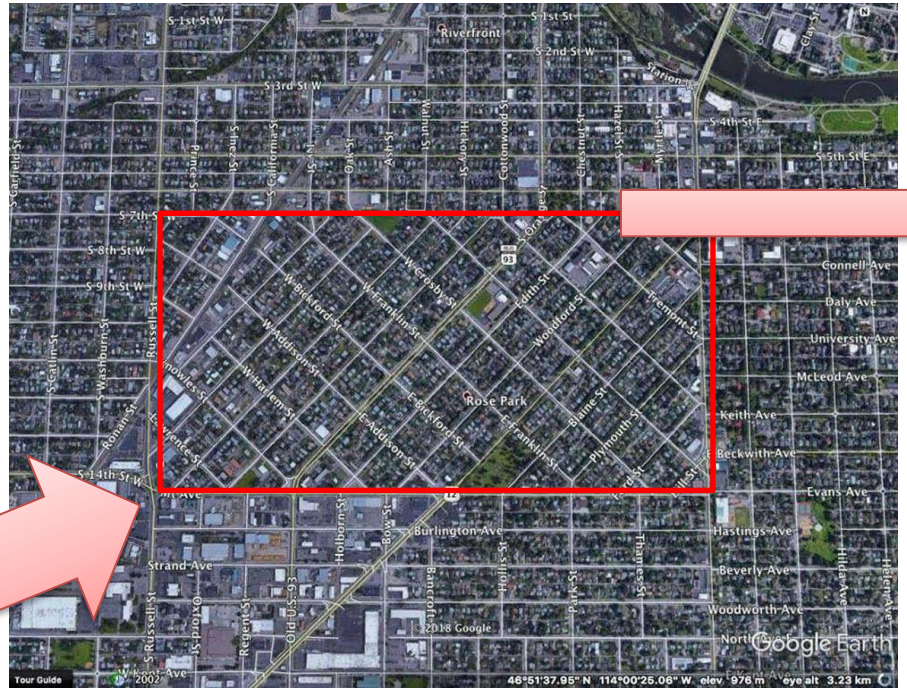


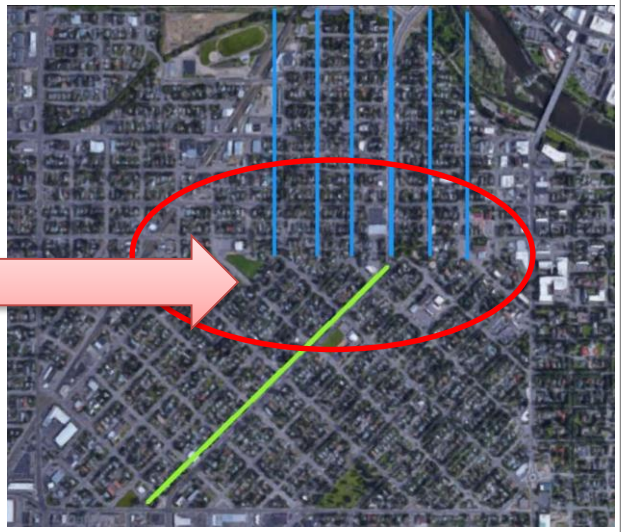
Source: (Anthony J. Richardson et. al, 1995)

Examples of Inadequate Transport Planning

Lack of City Planning

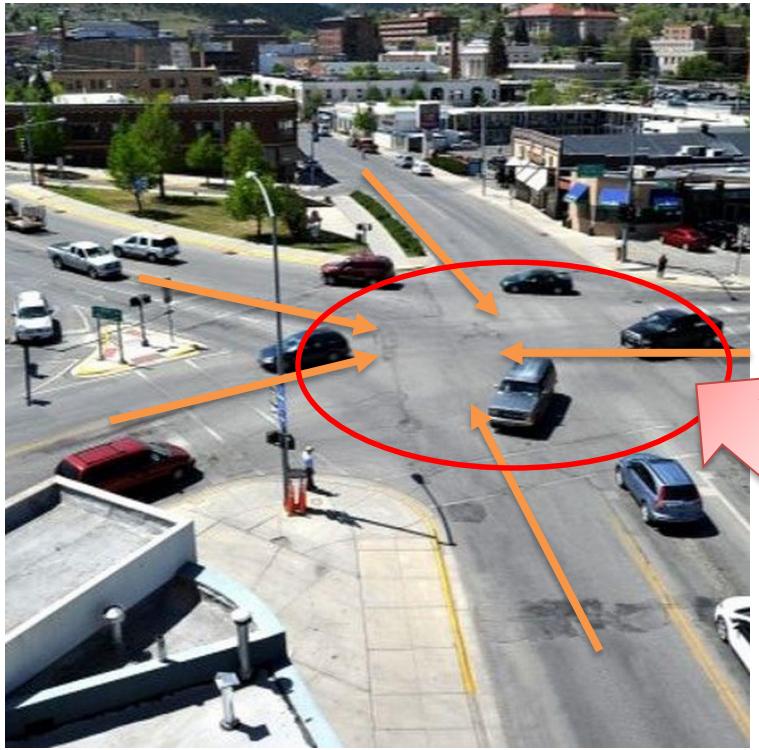
Example of the City of Missoula





Source: (https://www.re-thinkingthefuture.com/architects-lounge/a1517-10-examples-of-bad-urban-city-planning/)

Example of Lack of City Planning in the City of Missoula



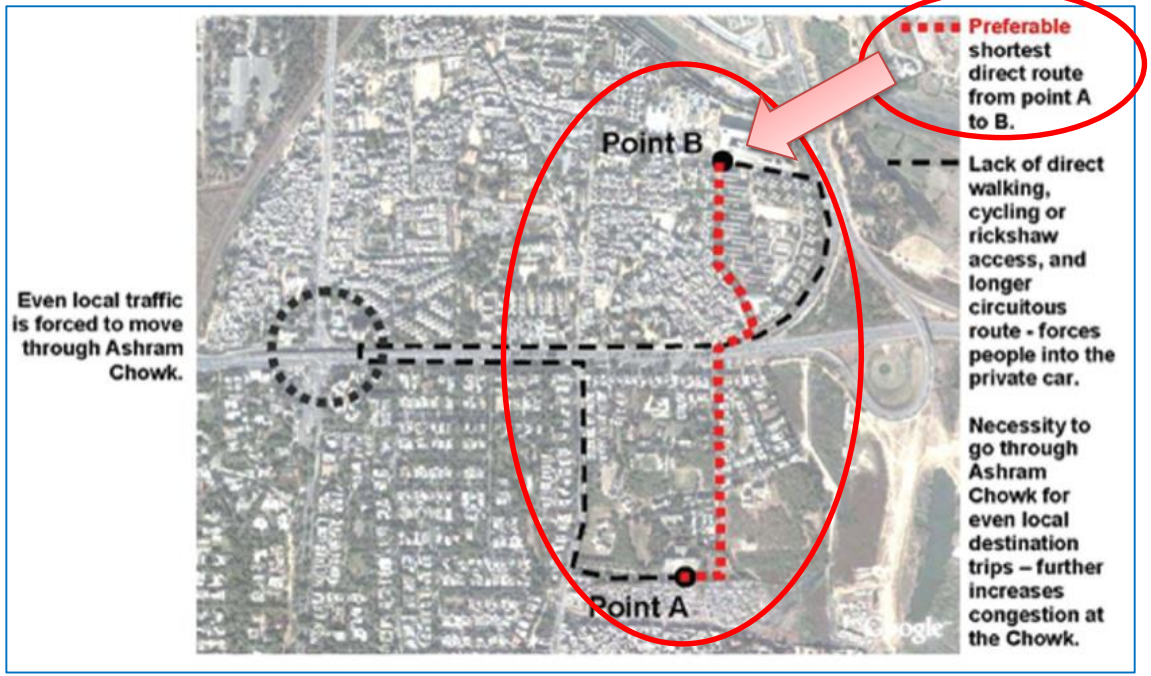
- Missoula is famous for its weird 'Slant streets' neighbourhood, named after the only section of the town that does not follow the usual grid pattern.
- Was a result of a clash of interest between the then developers, that led to construction of Complicated intersections in the city.

Clash of Intersections:
An example of the most dangerous "Malfunction Junction" in Missoula

Source: (https://www.re-thinkingthefuture.com/architects-lounge/a1517-10-examples-of-bad-urban-city-planning/)

Consequences of Inadequate Transport Planning?

Vehicular Congestion



- Increased congestion at Ashram Chowk Ring Road, New Delhi due to Closure of secondary networks

Source: (MOUD, 2016)

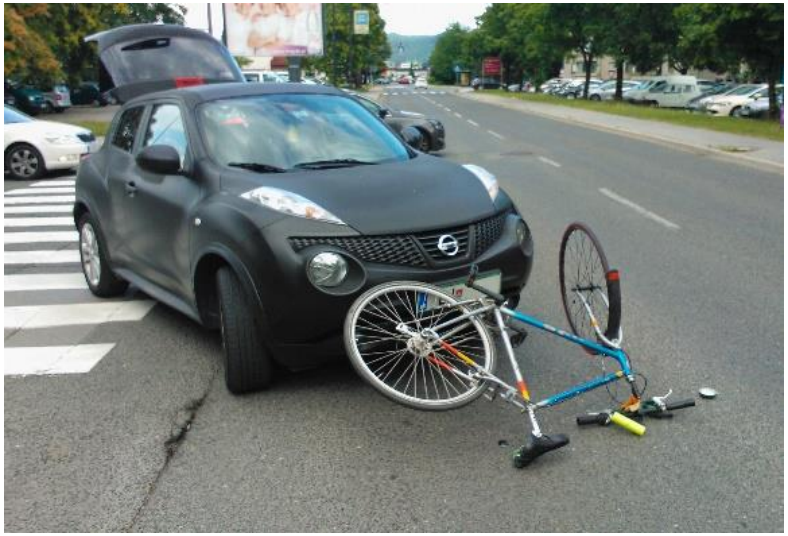
Congestion at Terminals and Transport modes





Accidents





Source: (https://www.re-thinkingthefuture.com/architects-lounge/a1517-10-examples-of-bad-urban-city-planning/)

Source: (https://www.reliance-foundry.com/blog/flex-post-bike-lane)

Traffic emissions



- More the congestion and idling, more pollutant emissions

Source: (https://www.re-thinkingthefuture.com/architects-lounge/a1517-10-examples-of-bad-urban-city-planning/)

Poor Parking facilities





Source: (MOUD, 2016)

Example of poor parking in Delhi street is due to poor planning

Source: (https://www.re-thinkingthefuture.com/architects-lounge/a1517-10-examples-of-bad-urban-city-planning/)

Poor Non-Motorized Transport facilities

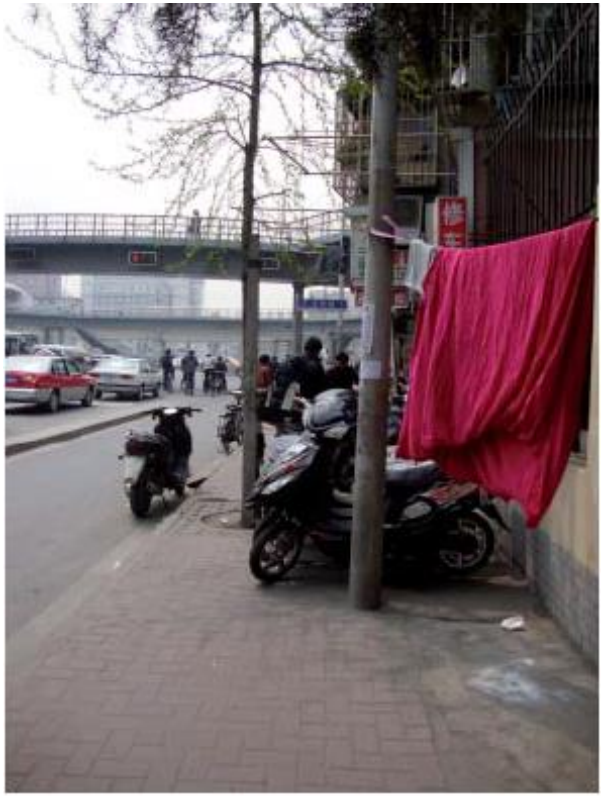


- Dangerous for Pedestrians and bicycles



Source: (https://www.mid-day.com/mumbai/mumbai-news/article/Finally--Thane-stadium-opens-for-cricketers-15765189?infinite-scroll=3)

Unsafe Pedestrian paths



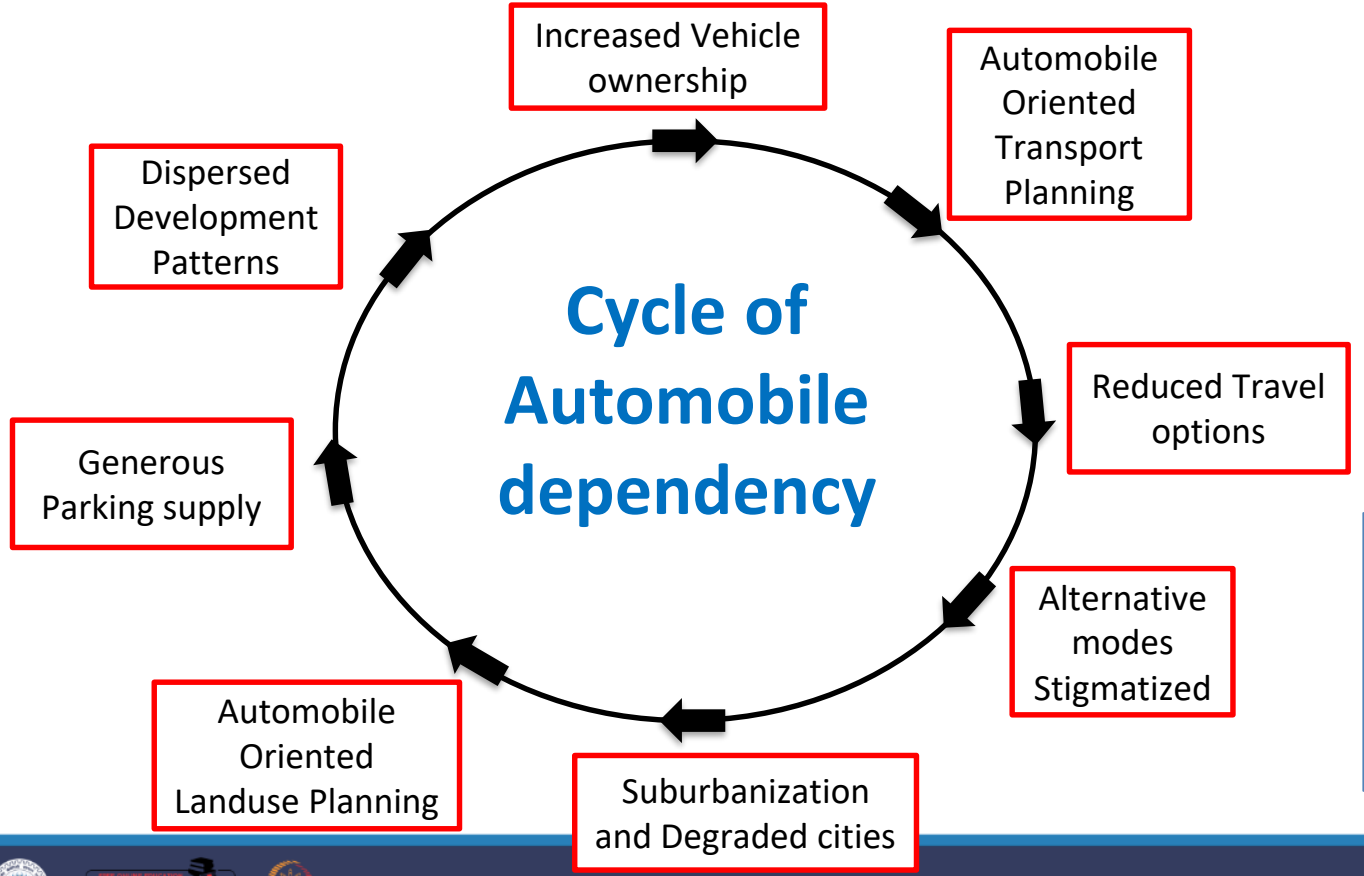
Source: (Metrasys, 2012)



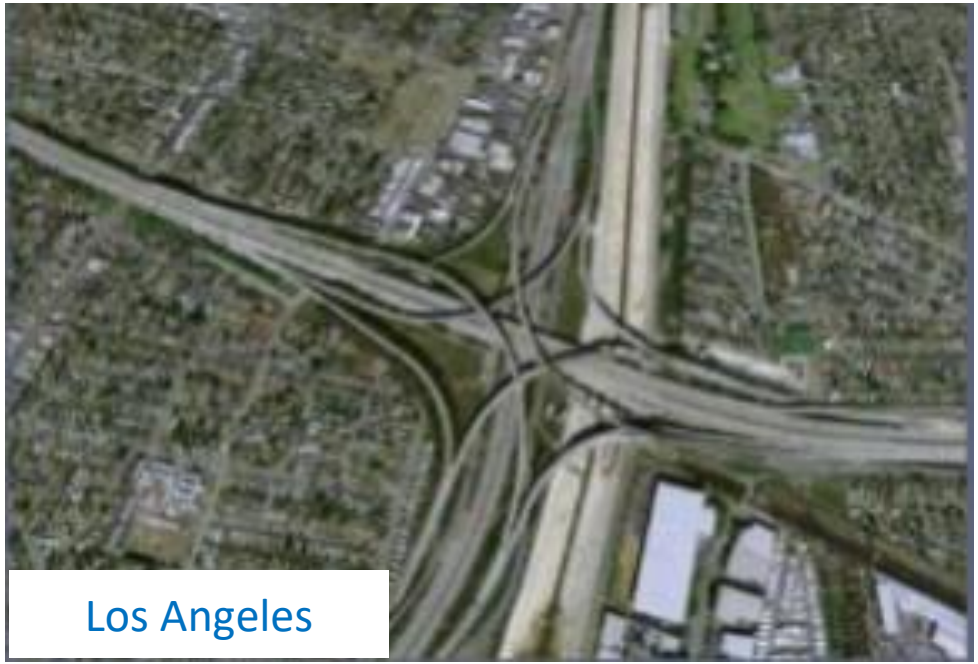
Source: (https://www.mid-day.com/mumbai/mumbai-news/article/Finally--Thane-stadium-opens-for-cricketers-15765189?infinite-scroll=3)

The Cycle of Automobile dependency

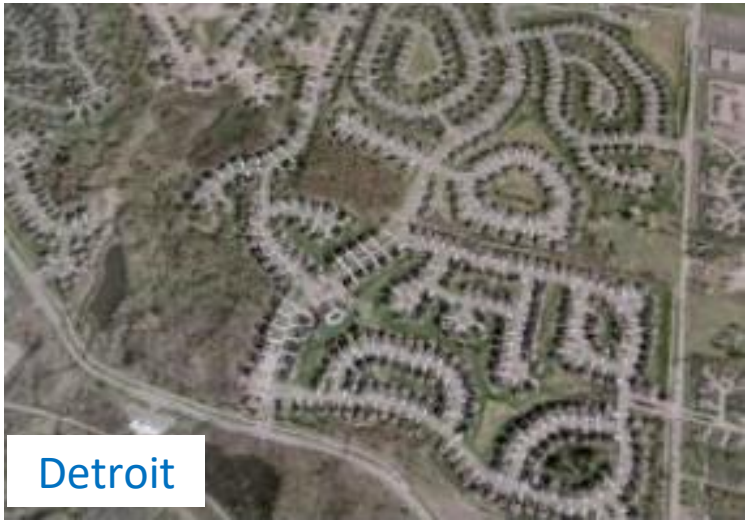




Example of the North American Car-dependent Model

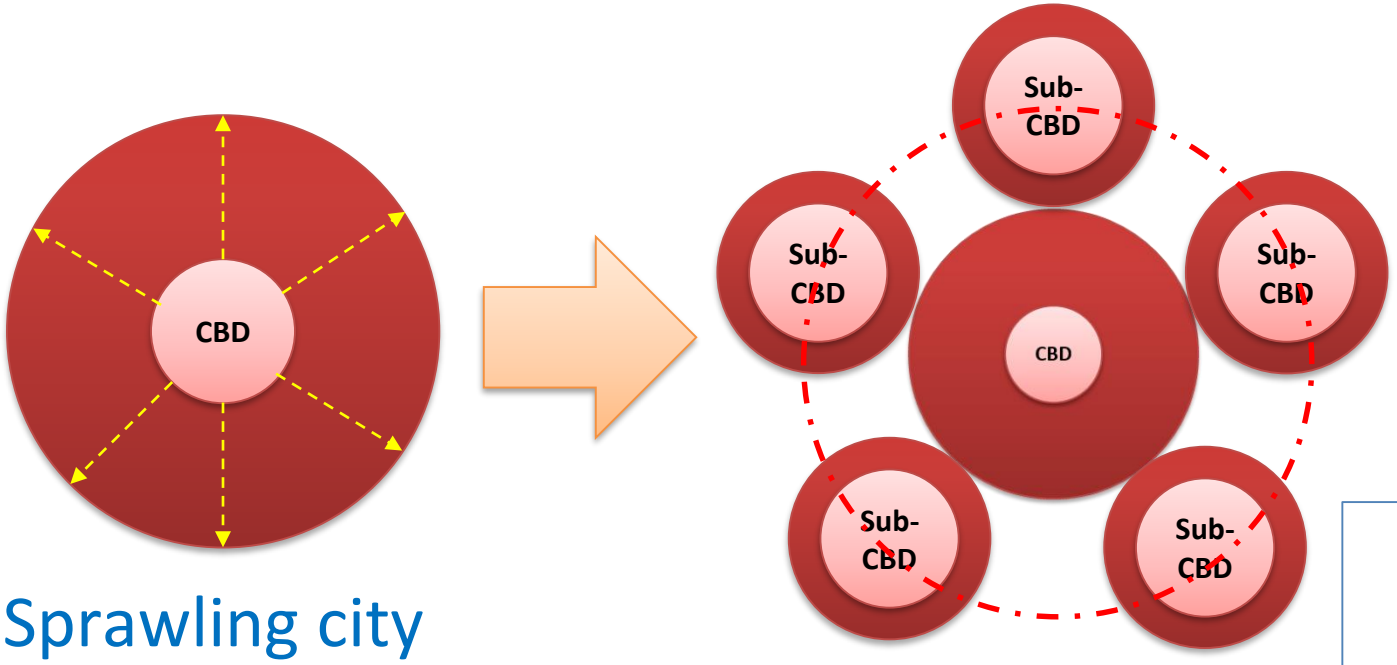


Los Angeles



Detroit

From Sprawling cities to Multi-polar centralized cities

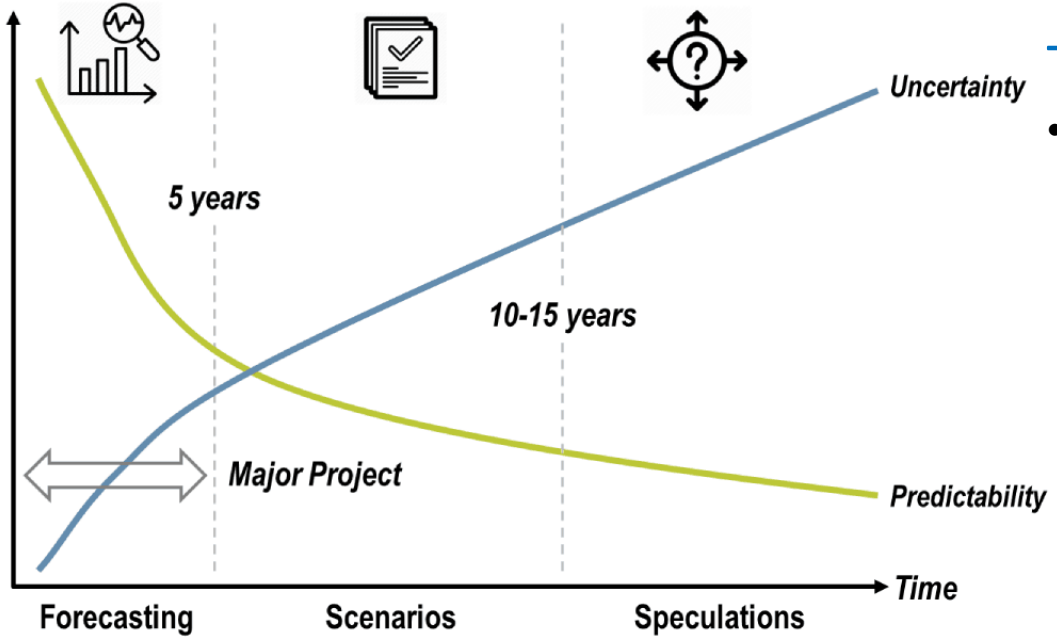


Sprawling city

Multi-polar centralized cities

CBD: Central Business District or commonly known as city center

Transport Planning based on Induced Demand

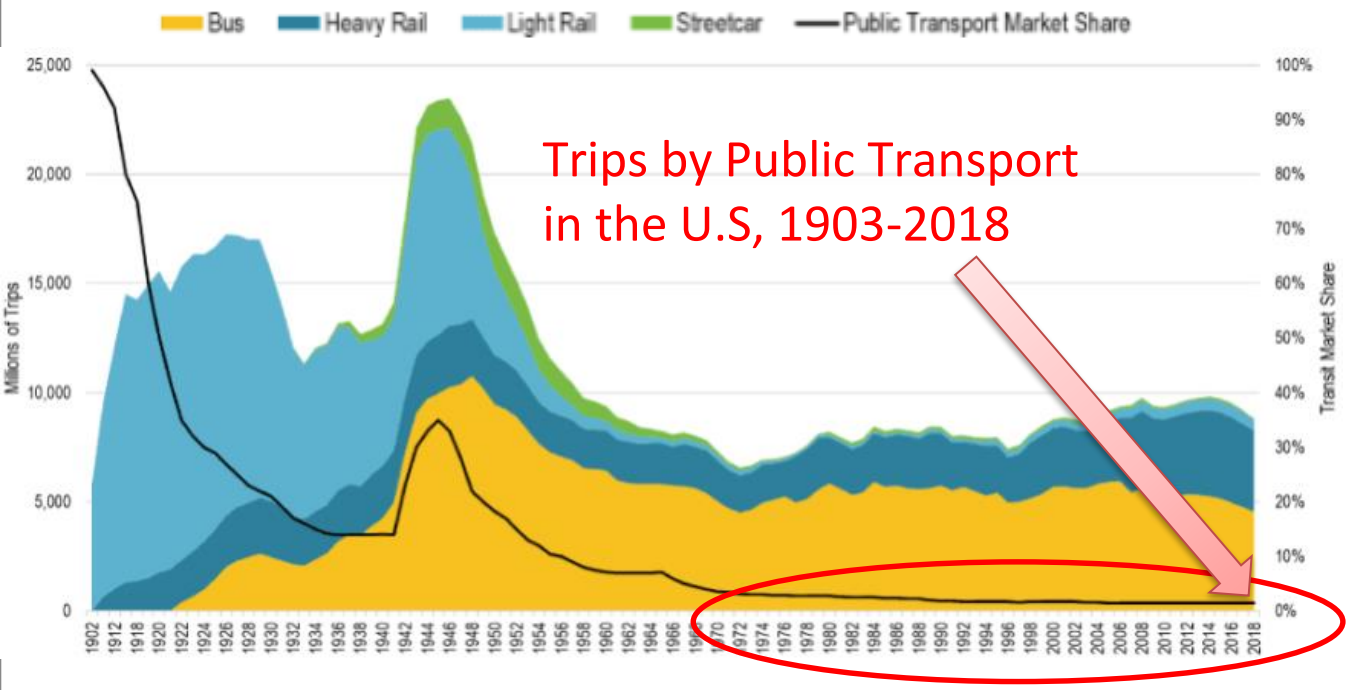


Transport Planning in 1940-80s: To predict and accommodate

- The most common prediction of the modeling is that present capacities will be unable to cope with traffic growth, the tendency has been to produce **planning solutions that call for an expansion of capacity**

Source: (Jean-Paul Rodrigue 2020)

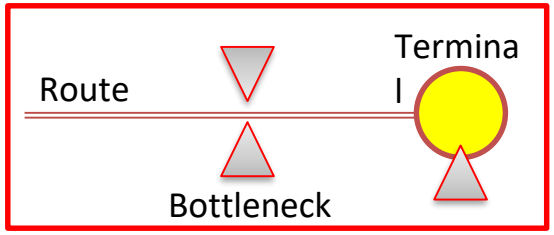
Example of Transport Planning in the United States



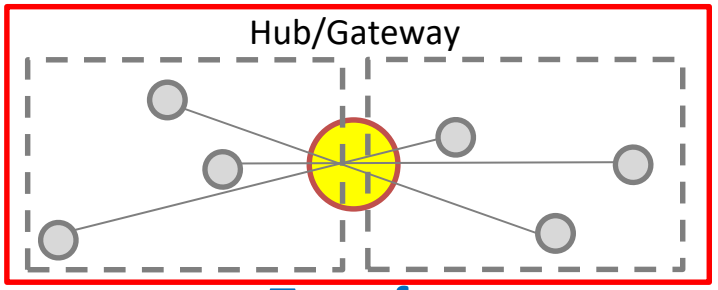
Individual Mobility and Motorization substantially decreased the Urban mobility in Public transit to less than 2% in the U.S.

Source: (Jean-Paul Rodrigue 2020)

Common Problems in Transport Planning

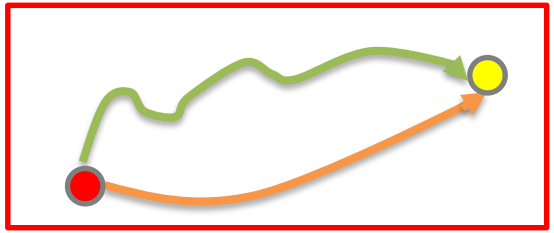


Capacity

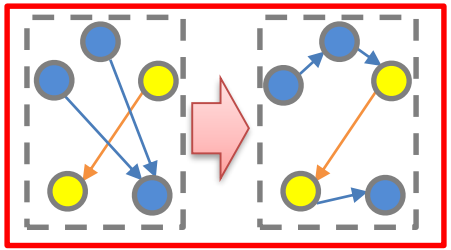


Transfe

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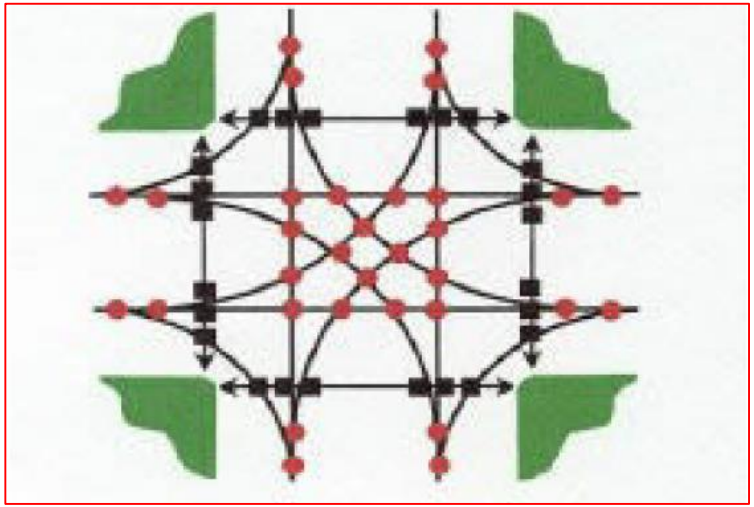
Reliability



Integratio

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Vehicle conflicts at Conventional Intersections



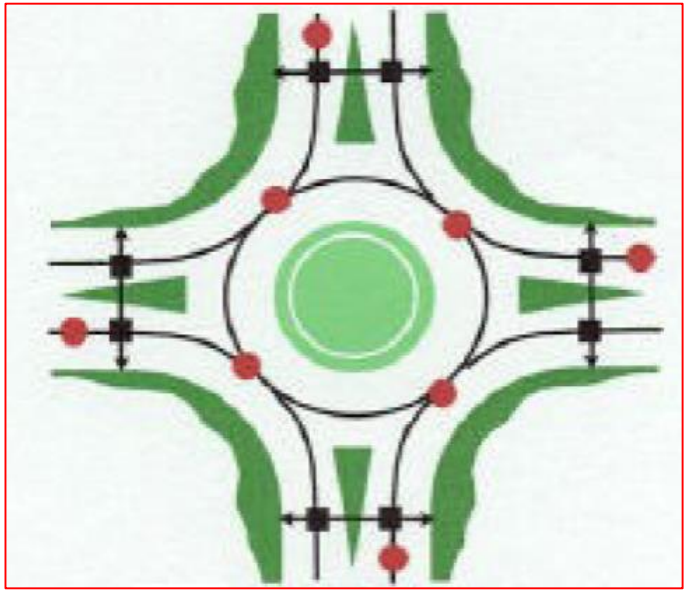
32 Vehicle to vehicle conflicts

24 Vehicle to pedestrian conflicts

Now, What are the Planning Solutions?



Example of Reduced Vehicle conflicts at Roundabouts

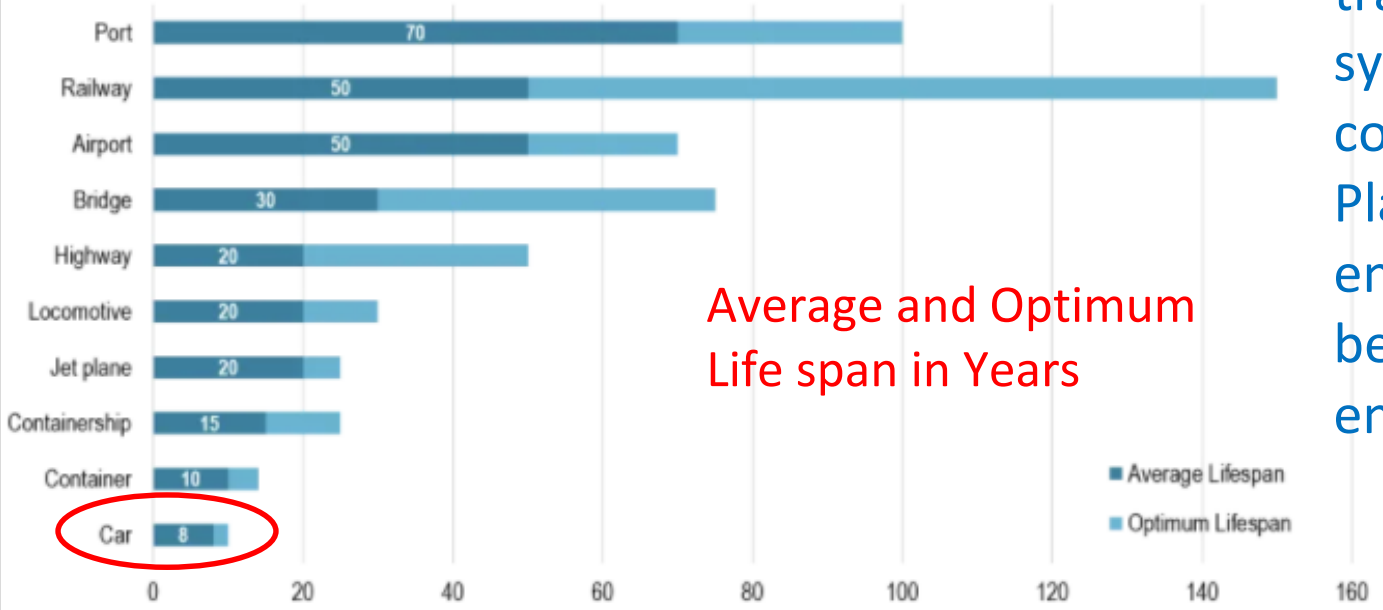


Conflict points:

8 Vehicle to vehicle

8 Vehicle to pedestrian

Considering the Average Life-span of Transport Assets in Planning



Average and Optimum Life span in Years

The life span of the transportation systems should be considered in the Planning process to ensure Sustainable benefits to environment

Source: (Jean-Paul Rodrigue 2020)

Efficient Street design







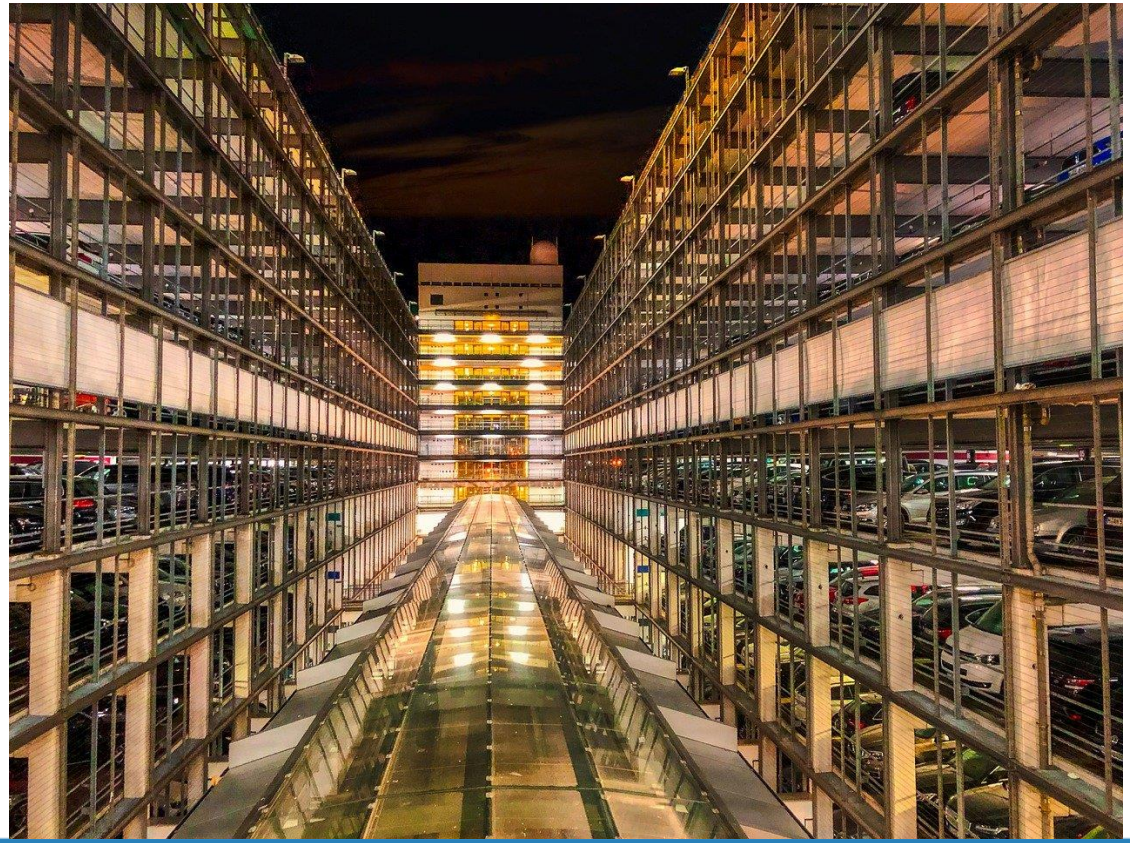






Source: (Metrasys, 2012)

Efficient Parking Facilities





Segregated Bus lanes







Source: (Metrasys, 2012)

Quality Bus stops



A boy sits on the swing and reads one of the books available at Singapore's bus stop.

Source: (https://www.bloomberg.com/news/articles/2017-03-01/singapore-may-have-designed-the-world-s-best-bus-stop)

Accessibility for Specially-abled



Source: (Metrasys, 2012)

Park and Ride facilities





Safe Pedestrian Zones











Source: (Metrasys, 2012)

Shared spaces with Proper planning



Safe Pedestrian crossings on Streets



Narrowing of roads



Zebra crossing with pedestrian island



Pedestrian bridge

Crossings with traffic light



Source: (Metrasys, 2012)

Uninterrupted Bicycle paths



The 25 km long uninterrupted Bike path in Turkey

Source: (https://www.trtworld.com/life/turkey-opens-world-s-longest-uninterrupted-bike-path-38786)

Cycle lanes at Intersections and Bus stops





Cycle lanes at Intersections

Cycle lanes at Bus stops

Bicycle parking facilities





Multistorey bicycle parking at Amsterdam

Real time Information for People





Source: (Metrasys, 2012)

Conclusion

- Obsolete design of Transportation systems and infrastructure are the major reasons leading to negative externalities such as accidents and traffic fatalities.

- Proper transportation planning implemented with the right management strategies and policy intervention can lead to sustainable transport planning solutions.



References

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- The City Alliance, (2009). "*Traffic Intersection Points of Conflict*", Graphic depiction of vehicle and pedestrian conflicts at signalized intersections vs roundabouts.

Thank You